

# FORCE and MOTION: Modeling Falling and Skydiving

CS 313 – Computational Science  
1<sup>st</sup> Sem., S.Y. 2022 - 2023



# Introduction

In this lesson, a system dynamics tool will be used to model the motion of someone skydiving. Such a jump has two phases, a free-fall stage followed by a parachute stage with greater air friction.



# Acceleration, Velocity and Position

- The instantaneous rate of change, or derivative, of position (s) with respect to time (t) is **velocity (v)**.
- The instantaneous rate of change of velocity with respect to time is **acceleration (a)**. In derivative notation,

$$v(t) = \frac{ds}{dt}$$

$$a(t) = \frac{dv}{dt}$$



# Acceleration, Velocity and Position

## Formulating a model for a falling object

a. Four variables for the model and their units in the metric system.

- ❖ time, perhaps in s;
- ❖ distance, perhaps in ms;
- ❖ velocity, perhaps in m/s;
- ❖ acceleration, perhaps in m/s<sup>2</sup>

b. Differential equation relating time (t), position (s), and velocity (v).

$$v(t) = \frac{ds}{dt}$$

c. Differential equation relating time (t), velocity (v), and acceleration (a).

$$a(t) = \frac{dv}{dt}$$



# Acceleration, Velocity and Position

## Formulating a model for a falling object

d. Ignoring friction, give any of the following that are constant in a fall: time, distance, velocity, acceleration.

e. In a model diagram, list the components that will be in stocks (box variables):  
 $t, s, v, a, ds/dt, dv/dt$ .

f. In a model diagram, give the value(s) that will flow into the position stock (box variable) for change in position:  $t, s, v, a, ds/dt, dv/dt$ .

g. In a model diagram, give the value(s) that will flow into the velocity stock (box variable) for change in velocity:  $t, s, v, a, ds/dt, dv/dt$ .



# Acceleration, Velocity and Position

## Formulating a model for a falling object

d. Ignoring friction, give any of the following that are constant in a fall: time, distance, velocity, acceleration.

→ Acceleration due to gravity,  $9.81 \text{ m/s}^2$

e. In a model diagram, list the components that will be in stocks (box variables):

$t, s, v, a, ds/dt, dv/dt$ .

→  $s$  and  $v$

f. In a model diagram, give the value(s) that will flow into the position stock (box variable) for change in position:  $t, s, v, a, ds/dt, dv/dt$ .

→  $ds/dt$

g. In a model diagram, give the value(s) that will flow into the velocity stock (box variable) for change in velocity:  $t, s, v, a, ds/dt, dv/dt$ .

→  $dv/dt$  or  $a$ , which is the constant acceleration due to gravity without friction



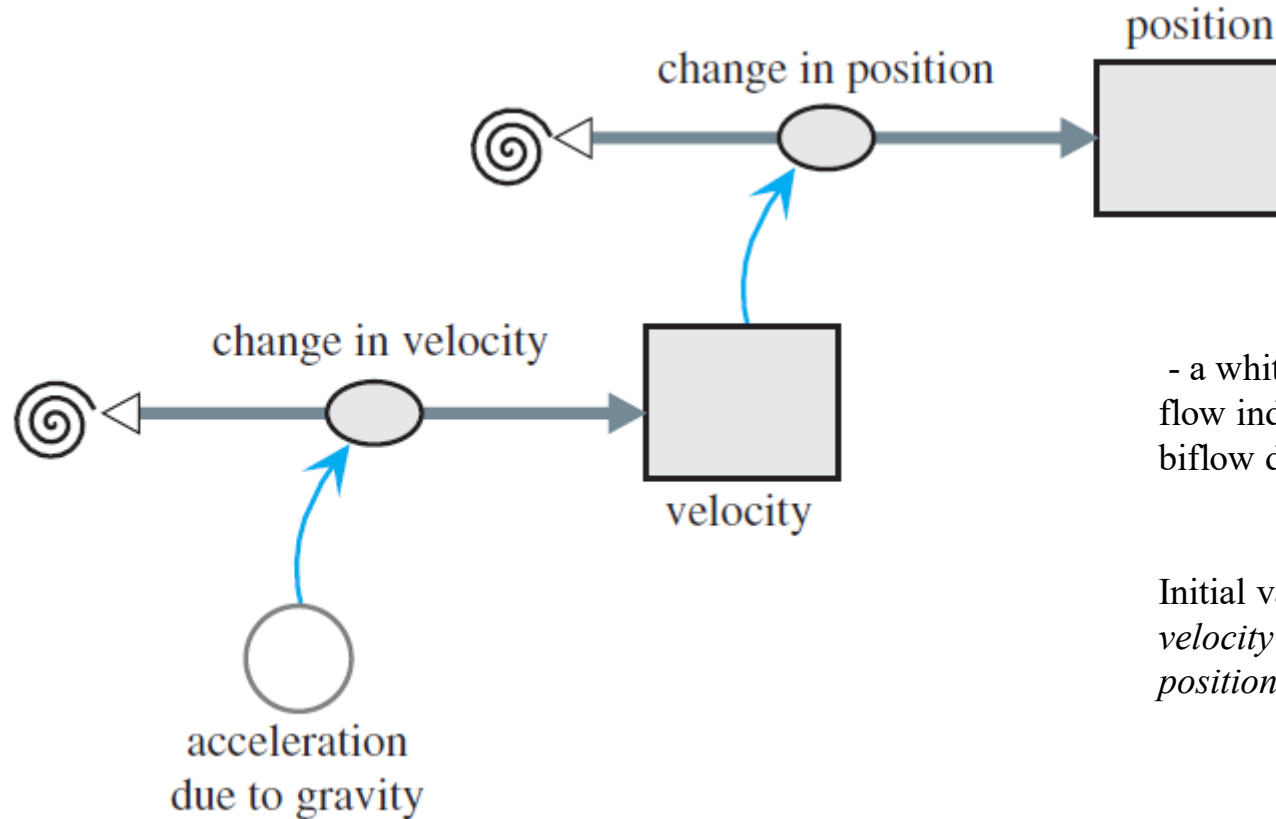
# Acceleration, Velocity and Position

## Formulating a model for a falling object

**Stocks** (box variables) for the quantities that accumulate, the height (*position*) and velocity (*velocity*) of the ball.

A **flow** representing the change goes into velocity (*change\_in\_velocity*) and a flow for the change in height (*change\_in\_position*).

A **converter/variable** (*acceleration\_due\_to\_gravity*) contains the constant for *acceleration due to gravity*, which is approximately  $-9.81 \text{ m/s}^2$



- a white arrowhead on each flow indicates secondary biflow direction.

Initial values:  
*velocity* = 15 m/s  
*position* = 11 m

Diagram of motion of ball thrown straight up

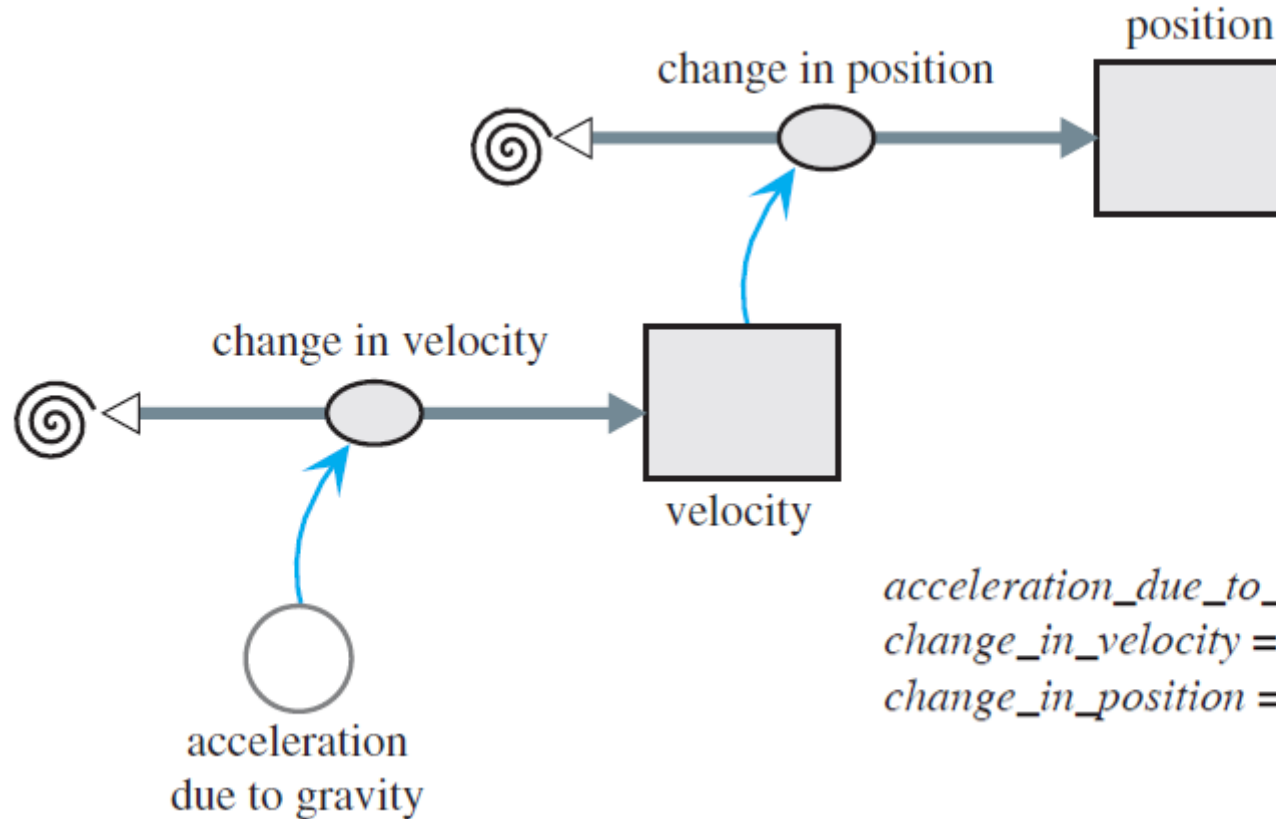
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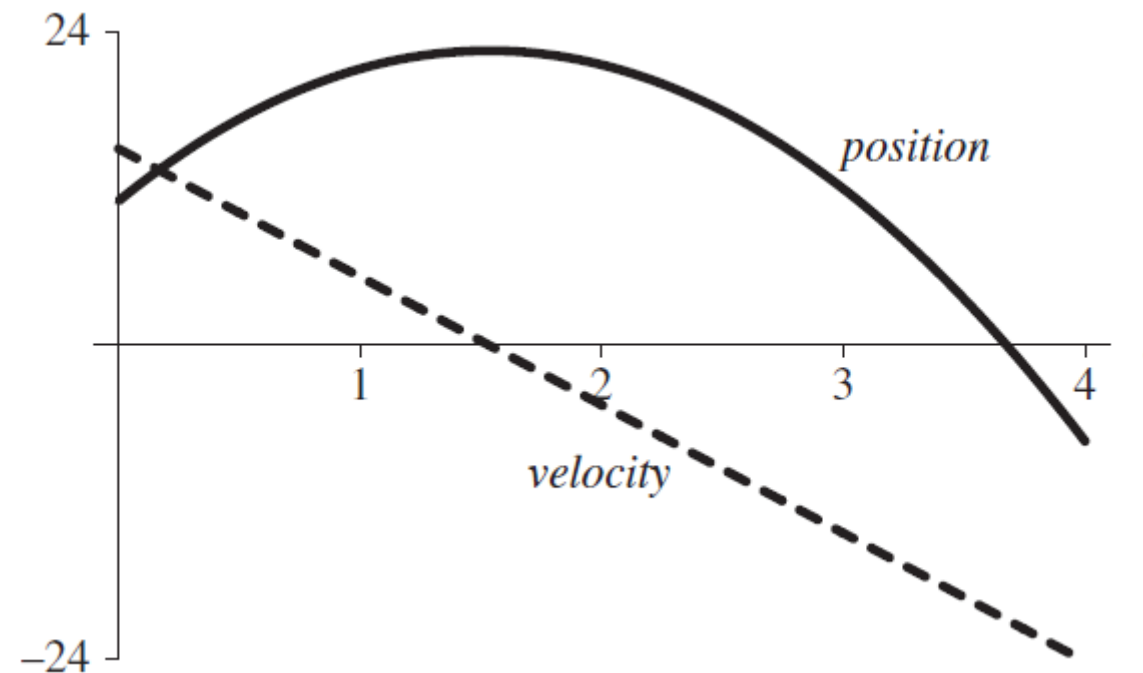


$$\begin{aligned} \text{acceleration\_due\_to\_gravity} &= -9.81 \text{ m/s}^2 \\ \text{change\_in\_velocity} &= \text{acceleration\_due\_to\_gravity} \\ \text{change\_in\_position} &= \text{velocity} \end{aligned}$$

Diagram of motion of ball thrown straight up

# Acceleration, Velocity and Position

Formulating a model for a falling object

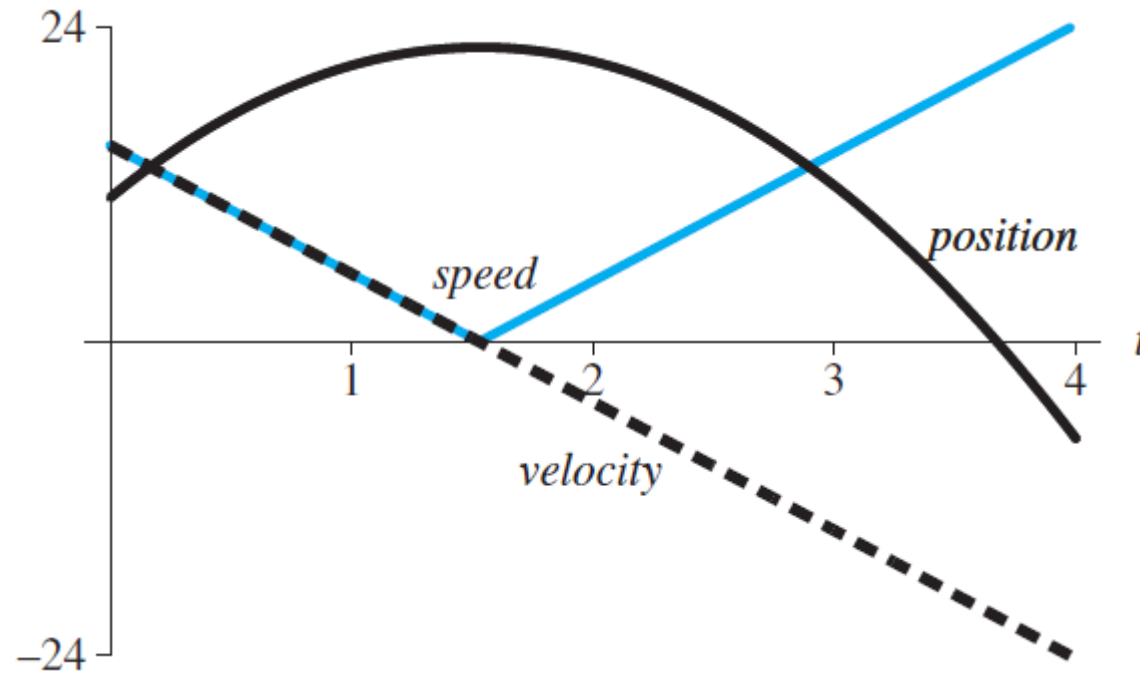


Graph of velocity (m/s) and position (m) of ball versus time (s)

# Acceleration, Velocity and Position

## Formulating a model for a falling object

For some of the models, it is more convenient to consider speed than velocity. The speed gives the magnitude of the change in position with respect to time, while the velocity expresses the magnitude with the direction. Thus, speed is the absolute value of the velocity.



Graph of velocity (m/s), position (m), and speed (m/s) of ball versus time (s)

# Physics Background

**Newton's second law** A force  $F$  acting on a body of mass  $m$  gives the body acceleration  $a$  according to the following formula:

$$F = ma$$

**Definition** A newton (N) is a measure of force, and  $1 \text{ N} = 1 \text{ kg m/s}^2$ .

- The acceleration is directly proportional to the force and inversely proportional to the mass:
- Weight is a force and is not the same as mass. The acceleration involved is acceleration due to gravity, which at sea level is about  $-9.81 \text{ m/s}^2$  or  $-32 \text{ ft/s}^2$  for up being the positive direction.

For example, an object that has mass of 20 kg has a weight of  $-196.2$  newtons (N), as the following shows:

$$\text{weight} = F = (20 \text{ kg})(-9.81 \text{ m/s}^2) = -196.2 \text{ kg m/s}^2 = -196.2 \text{ N}$$



# Physics Background

Kinetic friction, or drag, also is a force. This force between objects is in the opposite direction to a moving object and tends to slow motion. Thus, *kinetic friction dampens motion of an object.*

For a small object traveling slowly, such as a dust particle floating through the air, **Stokes's friction** is usually employed, which states that *friction on the particle is approximately proportional to its velocity,*

$$\mathbf{F} = k\mathbf{v}$$

where  $k$  (kg/s) is a constant of proportionality for the particular object and fluid and  $v$  (m/s) is the velocity.



# Physics Background

For a larger object moving faster through a fluid, we usually employ **Newtonian friction**, which states that the drag is approximately as follows:

$$F = 0.5CDAv^2$$

where

*C - a dimensionless constant of proportionality (the coefficient of drag, or drag coefficient) related to the shape of the object,*

*D - the density of the fluid, and*

*A - the object's projected area in the direction of movement.*



# Physics Background

- At 0 °C, the density of air at sea level is 1.29 kg/m<sup>3</sup>.
- For shapes that are hydrodynamically good,  $C < 1$ ;
- For spheres,  $C$  is about 1; and
- For shapes that are hydrodynamically inefficient,  $C > 1$ .
- Many objects have a coefficient of drag of about 1.

Through air with  $C = 1$ , Newtonian friction is approximately the following:

$$F = 0.65Av^2$$

*The density of water at 3.98 °C, where the fluid achieves its maximum density, is 1.00000 g/cm<sup>3</sup>.*



# Physics Background

- The drag force is in the opposite direction of motion, and the sign of velocity indicates the direction.
- On the upward portion of a trajectory, drag and gravity both act downward; while on the downward part, drag is upward and gravity downward.
- For the general formula for Newtonian friction, we take the absolute value of only one of the velocity terms and multiply the entire formula by  $-1$ , yielding  $-0.5CDAv|v|$ .

*If ABS is the absolute value function, the translation of this formula into a system dynamics tool is as follows:*

$$-0.5 * \text{drag\_coefficient} * \text{density} * \text{projected\_area} * \text{velocity} * \text{ABS}(\text{velocity})$$



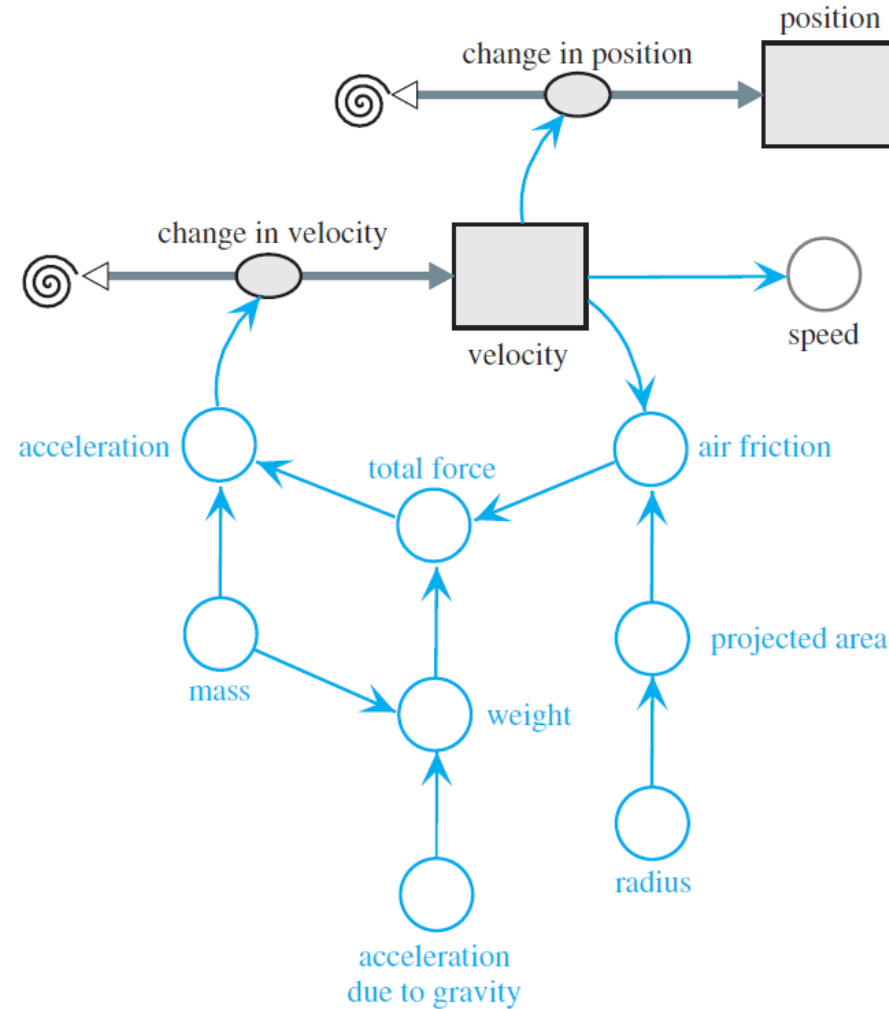
# Summary of Several Models for Magnitude of Fluid Friction

| <i>Name</i>                    | <i>Formula</i>  | <i>Meanings of Symbols</i>                                                                                              | <i>When to Use</i>                                              |
|--------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Stokes's friction              | $F = kv$        | $k$ constant<br>$v$ velocity                                                                                            | Very small object moving slowly through fluid                   |
| Newtonian friction             | $F = 0.5CDAv^2$ | $C$ coefficient of drag<br>$D$ density of fluid<br>$A$ object's projected area in direction of movement<br>$v$ velocity | Larger objects moving faster through fluid                      |
| Newtonian friction through air | $F = 0.65Av^2$  | $A$ object's projected area in direction of movement<br>$v$ velocity                                                    | Larger objects with $C = 1$ moving faster through sea-level air |



# Friction during Fall

Two forces on the ball are considered, gravity and drag friction. The force due to gravity is its weight, which by Newton's second law is  $F = ma$ .

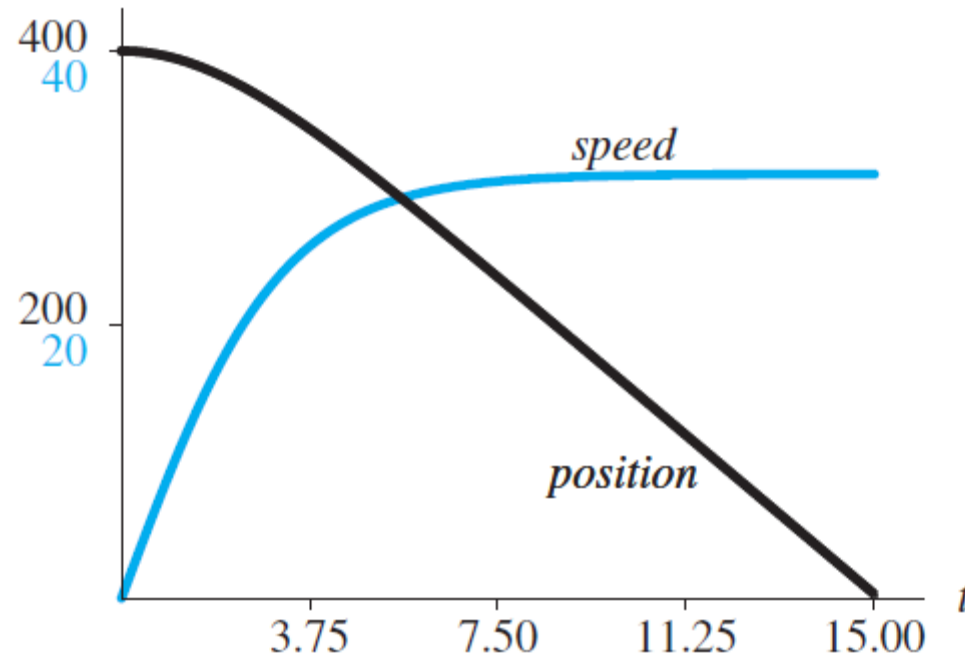


$mass = 0.5 \text{ kg}$   
 $acceleration\_due\_to\_gravity = -9.81 \text{ m/s}^2$   
 $radius = 0.05 \text{ m}$   
 $weight = mass * acceleration\_due\_to\_gravity$   
 $projected\_area = 3.14159 * radius^2$   
 $air\_friction = -0.65 * projected\_area * velocity * ABS(velocity)$   
 $total\_force = weight + air\_friction$   
 $acceleration = total\_force / mass$   
 $change\_in\_velocity = acceleration$   
 $change\_in\_position = velocity$   
 $speed = ABS(velocity)$   
 $velocity(0) = 0 \text{ m/s}$   
 $velocity(t) = velocity(t - \Delta t) + (change\_in\_velocity) * \Delta t$   
 $position(0) = 400 \text{ m}$   
 $position(t) = position(t - \Delta t) + (change\_in\_position) * \Delta t$

Diagram for motion of ball under influence of air friction



# Friction during Fall



Graph of position (m) and speed (m/s) of object versus time (s) under influence of friction

# Modeling a Skydive

## \*Laboratory Activity

